PassFrame tuition

Number Sequences

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 5 (+1 each time)

2. 25 (+5 each time)

3. 10 (+2 each time)

4. 12 (−2 each time)

5. 15 (+3 each time)

6. 60 (−10 each time)

7. 20 (+4 each time)

8. 30 (−5 each time)

9. 32 (×2 each time)

10. 9 (+2 each time — odd numbers)

11. 12 (+2 each time)

12. 45 (−9 each time)

13. 30 (+6 each time)

14. 35 (+7 each time)

15. 50 (−10 each time)

Standard

1. 64 (square numbers: 4^2, 5^2, 6^2, 7^2, next 8^2)

2. 192 (×2 each time)

3. 768 (×4 each time)

4. 12 (÷4 each time)

5. 55 (triangular numbers: T6, T7, T8, T9 (gaps 7, 8, 9, 10), next gap 10: 45+10)

6. 50 (gaps are consecutive odd numbers: +5, +7, +9, +11, next +13)

7. 224 (×2 each time)

8. 16 (÷2 each time)

9. 24 (gaps grow by 1 each time: +1, +2, +3, +4, +5, next +6)

10. 972 (×3 each time)

11. 20 (−9 each time)

12. 55 (each number is the sum of the two before it (Fibonacci): 21+34)

13. 144 (square numbers: 8^2, 9^2, 10^2, 11^2, next 12^2)

14. 1458 (×3 each time)

15. 16 (−6 each time)

Standard / Hard

1. 21 (alternates +5, −2: 16 + 5 = 21)

2. 6 (two patterns woven together: 1st/3rd/5th terms 1,3,5 go +2; 2nd/4th terms 10,8 go −2, so next even term is 8−2=6)

3. 29 (alternates +3, ×2: 26 + 3 = 29)

4. 27 (gaps are odd numbers: +1, +3, +5, +7, next +9: 18+9=27)

5. 13 (Fibonacci sequence: each number is the sum of the two before it)

6. 33 (add the digit sum of the number each time: 30 + (3+0) = 33)

7. 16 (two patterns woven together: 1st/3rd/5th terms 3,9,27 go ×3; 2nd/4th terms 4,8 go ×2, so next even term is 8×2=16)

8. 4 (odd-position terms 1,2,3 go +1; even-position terms 4,7,10 go +3, so the next odd-position term is 3+1=4)

9. 35 (each term is a square number minus 1: 1^2-1, 2^2-1, 3^2-1… next is $6^2-1=35$)

10. 98 (each number is doubled, then 2 is subtracted: 50×2−2=98)

11. 125 (cube numbers: 1^3, 2^3, 3^3, 4^3, 5^3)

12. 25 (gaps are odd numbers getting bigger: −1, −3, −5, −7, next −9: 34−9=25)

13. 38 (alternates +1, ×2: 19 × 2 = 38)

14. 14 (each number is the sum of the two before it, minus 1: 6+9−1=14)

15. 78 (gaps grow by 4 each time: +7, +11, +15, +19, next +23: 55+23=78)

Homework — Answers

Easy / Standard

1. 7 (+1 each time)

2. 50 (+10 each time)

3. 13 (+3 each time)

4. 18 (−3 each time)

5. 40 (+8 each time)

6. 36 (−6 each time)

7. 96 (×2 each time)

8. 25 (−5 each time)

9. 324 (×3 each time)

10. 55 (+11 each time)

11. 36 (−12 each time)

12. 45 (+9 each time)

13. 17 (+4 each time)

14. 35 (−7 each time)

15. 26 (+5 each time)

Standard

1. 81 (square numbers: 5^2, 6^2, 7^2, 8^2, next 9^2)

2. 256 (×2 each time)

3. 512 (×4 each time)

4. 16 (÷4 each time)

5. 66 (triangular numbers: T7, T8, T9, T10 (gaps 8, 9, 10, 11), next gap 11: 55+11)

6. 58 (gaps are consecutive odd numbers: +7, +9, +11, +13, next +15)

7. 288 (×2 each time)

8. 20 (÷2 each time)

9. 25 (gaps grow by 1 each time: +1, +2, +3, +4, +5, next +6)

10. 1215 (×3 each time)

11. 26 (−11 each time)

12. 89 (each number is the sum of the two before it (Fibonacci): 34+55)

13. 169 (square numbers: 9^2, 10^2, 11^2, 12^2, next 13^2)

14. 1701 (×3 each time)

15. 22 (−7 each time)

Standard / Advanced

1. 11 (two patterns woven together: odd-position terms 6,7,8 go +1; even-position terms 9,10 go +1, so next even term is 10+1=11)

2. 16 (the gap grows by 1 each time: +2, +3, +4, next +5)

3. 18 (alternates ×2, −1: 9 × 2 = 18)

4. 66 (the gap shrinks by 1 each time: −10, −9, −8, next −7)

5. 26 (the gaps are 1, 2, 4, 7 — and the gaps between THOSE gaps grow by 1 each time, so the next gap is 11, giving 15+11=26)

6. 62 (49 + (4+9) = 49 + 13 = 62)

7. 35 (19 × 2 − 3 = 35)

8. 14 (this is an odd position; the odd-position numbers run 5, 8, 11, so the next is 11+3=14)

9. 15 (7 + 10 − 2 = 15)

10. 21 (15 + 6 = 21)

11. 202 (67 × 3 + 1 = 202)

12. 34 (the next gap is 10; 24 + 10 = 34)

13. 61 (54 + 7 = 61)

14. 38 (6^2 + 2 = 38)

15. 15 (24 − 9 = 15)